

respatial

Rotation notes: rigid repositioning of the stereo scene

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1 Motivation and mental model

Real recordings often carry a *global* spatial bias: the microphone placement or the balance decision shifts the entire scene to one side. The expand/contract warp (**--stage**, see the design notes) rescales the position distribution *about the center*; it can widen or narrow the stage, but it cannot move the bias itself. A right-biased recording, expanded, stays right-biased — and expands *more* into the right edge.

Rotation addresses exactly this. The mental model is *rotating the stereo microphone pair* (equivalently, rotating the stage around the listener): the whole scene turns rigidly. In position space this is a translation — the bias (median position) moves by θ , while the spread (the *p5..p95* width, the relative positions of all bins) is preserved. Rotation and expansion are therefore complementary: first rotate away the capture bias, then expand around the corrected center.

2 Position space and its inverse map

Deviation from center $d \in [-90^\circ, +90^\circ]$ (each channel spans 90° ; the total stage is up to 180°). The per-bin level ratio $r = |L|/|R| = 10^{\text{ild}/20}$ maps to deviation by

$$d = 2 \arctan(1/r) - 90^\circ, \quad \text{i.e.} \quad r(d) = \tan((90^\circ - d)/2), \quad (1)$$

so +ILD (L louder) maps to left ($d < 0$). This is the equal-power-panning convention: a source panned with $g_L = \cos \varphi$, $g_R = \sin \varphi$ has $r = \cot \varphi$ and $d = 90^\circ p$.

Rotation is a rigid translation in d -space:

$$d' = \text{clip}(d + \theta, \pm 89.9^\circ). \quad (2)$$

It is *not* a simple operation in ILD space, because ILD is a nonlinear, non-uniform function of d (a 1° shift near the center is a small ILD change, near the edge a large one). The implementation therefore works entirely in d -space and maps back through the inverse $r(d) = \tan((90^\circ - d)/2)$. The clip pins content pushed past the stage edge at the edge ($89.9^\circ \equiv \pm 61 \text{ dB ILD}$).

3 Unified energy-preserving gain

Given a target ratio r' for a bin of total amplitude $A = \sqrt{|L|^2 + |R|^2}$, the energy-preserving reweight is

$$|L'| = \frac{A r'}{\sqrt{1 + r'^2}}, \quad |R'| = \frac{A}{\sqrt{1 + r'^2}}, \quad \implies \quad |L'|^2 + |R'|^2 = A^2. \quad (3)$$

The per-channel gains $g_{L,R} = |L', R'|/|L, R|$ are real and non-negative, so *phase and ICPD are preserved exactly*; only the level ratio changes. This is the same gain form used by the warp (the warp is the $r' = r^\gamma$ special case); rotation, warp, and their composition share one code path.

4 Distance gain

Relocated content is level-adjusted as on a physical stage (listener at δ half-stage widths from the stage plane, `--depth`, default $\delta = 2$):

$$g_{\text{dist}}(d_0 \rightarrow d_1) = \frac{\sqrt{\delta^2 + (d_0/90^\circ)^2}}{\sqrt{\delta^2 + (d_1/90^\circ)^2}}. \quad (4)$$

It is applied equally to both channels, so it changes level but not position. For a $\pm 14^\circ$ rotation at $\delta = 2$ it is within $\pm 3\%$ of unity.

5 Composition with the warp

The order is fixed: *rotate first, then warp*,

$$d_{\text{rot}} = \text{clip}(d_0 + \theta, \pm 89.9^\circ), \quad \text{ild}_{\text{rot}} = \text{ild}(d_{\text{rot}}), \quad \text{ild}_{\text{final}} = \gamma \text{ild}_{\text{rot}}. \quad (5)$$

Rationale: correct the capture bias first, then expand about the corrected center. Warping first would amplify the bias asymmetrically — a right-biased distribution is pushed further right, with more clipping at the right edge. γ is solved by bisection so that $p95(|d_{\text{final}}|) = S/2$ for the requested stage width S (`--stage`); with no stage target, $\gamma = 1$ and the operation is a pure rotation.

6 Blend and RMS matching

`--blend` α crossfades original $\rightarrow$ final, $\text{out} = (1 - \alpha) \text{orig} + \alpha \text{final}$, which is a real non-negative per-bin gain ($g = (1 - \alpha) + \alpha g_{\text{final}}$): a pure amplitude interpolation, phase untouched. The output is scaled so its RMS matches the input (per-bin energy does not survive overlap-add), keeping A/B comparisons level-matched.

7 Realized vs. designed shift

The designed per-bin shift is exact in the *modified* spectrum, but the *measured* spectrum (re-analyzing the overlap-added signal) is the time-domain-consistent projection of it: the iSTFT/STFT round trip of an arbitrarily modified spectrum blends neighboring frames, so bins whose designed position changes rapidly from frame to frame are partially averaged. This is intrinsic to per-bin spectral modification (the warp has the same property).

Verification on the Mahler excerpt, $\theta = -14^\circ$:

- Identity path ($\theta = 0$, depth off): time-domain max error 4×10^{-6} ; per-bin $|\Delta d|$ $p95 = 4 \times 10^{-4}^\circ$ — the unmodified round trip is exact, as theory requires.
- On the common active-bin set (bins above the 5%-of-max gate in both original and output): median per-bin error $+0.8^\circ$, $p95 \approx 7^\circ$; median shift -13.4° vs. the nominal -14° .

- The overall measured median shift is -11.6° : the remaining $\approx 1.7^\circ$ comes from the active-bin set itself changing — the distance gain moves bin amplitudes by up to $\pm 3\%$, enough to push 7.2% of bins across the 5% gate, and the gate reweights the distribution.

For a mostly static spatial scene the projection error is small (sub-degree medians); it grows only where the per-frame position fluctuates.

8 Measurements and A/B results

Recording: Mahler, Symphony No. 7, mvt. 2 (44.1 kHz, 143.9 s).

- **Bias. measure** over the full file: median $d = +5.6^\circ$ (median ILD -0.85 dB) — the recording is right-biased. Segment 30–60 s: median $+14.4^\circ$, mean $+11.5^\circ$, $p5..p95 = -55.0^\circ.. +70.9^\circ$ (range 125.9°), $p95(|d|) = 72.8^\circ$, median ILD -2.20 dB.
- **Recentering** (`remix --rotate -14`): median $+14.4^\circ \rightarrow +2.8^\circ$ (shift -11.6° , Section 7), range $125.9^\circ \rightarrow 125.6^\circ$ (preserved, as a rigid shift requires), RMS ratio 1.000.
- **Recenter then expand** (`--rotate -14 --stage 180 --blend 0.5`): $\gamma = 21.1$, median $+11.7^\circ$, range 150.7° , RMS ratio 1.000.

A/B files: `report/mahler_30_60/{original_30_60, rotate14, rotate14_stage180_blend05}.wav`.

Commands:

```
python3 respatial.py measure samples/Mahler-7-2mvt.wav --t0 30 --t1 60
python3 respatial.py remix samples/Mahler-7-2mvt.wav --out r14.wav \
    --t0 30 --t1 60 --rotate -14
python3 respatial.py remix samples/Mahler-7-2mvt.wav --out r14s180.wav \
    --t0 30 --t1 60 --rotate -14 --stage 180 --blend 0.5
```

9 References

1. J. Blauert, *Spatial Hearing: The Psychophysics of Human Sound Localization*. Cambridge, MA: MIT Press, revised ed., 1997. The foundational treatment of the ILD/ITD localization cues and of microphone geometries; the basis of the d -space model used here. <https://mitpress.mit.edu/9780262021906/spatial-hearing/> (open-access copy: <https://direct.mit.edu/books/oa-monograph/4885/Spatial-HearingThe-Psychophysics-of-Human-Sound>).
2. V. Pulkki, “Virtual sound source positioning using vector base amplitude panning,” *J. Audio Eng. Soc.*, vol. 49, no. 10, Oct. 2001. Vector-based reformulation of amplitude panning; the equal-power pan law underlying the ILD $\leftrightarrow$ position map.
3. D. D. Lee and H. Seung, “Algorithms for non-negative matrix factorization,” in *Advances in Neural Information Processing Systems 13* (NIPS 2000). The NMF used by the detection stage (see the design notes). https://proceedings.neurips.cc/paper_files/paper/2000